FluxLink: A Unified Adaptive Platform for Secure Multi-Transport File Transfer

VE441 · App Development for Entrepreneurs · Project Proposal

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FluxLink — Elevator Pitch

Never waste time choosing how to send a file again —
FluxLink automatically selects the fastest, most secure transfer path across any platform and any network.

- **Zero decision cost** — routing is automatic
- **Any network** — LAN, P2P, relay fallback
- **Any platform** — Android, iOS, Desktop
- **Uniform E2EE** — all modes, all files
- **Encrypted chat** — context stays private
- **No account required** — pair via QR/token

Outline

Team Members

Team Name: NUUB

PM: Zhang Jinbin

Full Name	jAccount	Task Assignment	Technical Strengths
Sun Qizhen	qz_sun	Product architecture · Protocol routing engine · Network path scoring	Networking · Distributed systems · System design
Zhang Jinbin	moranxuege	PM · Mobile client (Android/iOS) · Cross-platform UI/UX	Android (Kotlin) · iOS (Swift) · UX implementation
Li Yuhao	y4954he	End-to-end encryption · QR/token handshake · Encrypted chat	Cryptography · Backend security · Protocol design
He Yumeng	17521218310	Relay server · Deployment · Testing · Cache policy	Backend · DevOps · QA · Django REST

Product & Research

Sun Qizhen
routing logic & protocol priority

Client & UX (PM)

Zhang Jinbin
mobile UI & cross-platform

Networking & Security

Li Yuhao
E2EE model & handshake

Backend & QA

He Yumeng
relay infra & test coverage

Interview Design — Participants & Venue

How We Find Interviewees

- SJTU engineering/CS communities
- GitHub discussions, V2EX, WeChat tech groups
- Screening: regularly transfer files across multiple devices/OS, and have encountered at least one of: size limits, incompatibility, network failure, privacy concerns
- **Avoid friends/family** — reduce social desirability bias

Venue & Session Structure

- Remote via Tencent Meeting / Zoom
- 10–20 min per session
- One interviewer + one note-taker
- Recorded with consent

0–5 min	Warm-up, device context
5–15 min	Pain stories, failure scenarios
15–20 min	Ideal workflow, retention

Theme 1: Device Context & Current Behavior

Goal: Establish devices/platforms used and current tools, without leading toward any product concept.

Q1 Walk me through the devices you use daily and how often they need to share files with each other.

Q2 Think back to the last few weeks — what triggered your need to send a file, and to whom?

Q3 What apps or methods do you normally reach for? How did you settle on those tools?

Q4 When you're about to send a file, what goes through your head before deciding which tool to use?

Theme 2: Pain Points & Failure Stories

Goal: Surface real breakdown moments, workarounds, and emotional friction.

Q5 Walk me through the last time you sent a large file — from the moment you decided to send it to delivery.

Q6 Has your usual method ever just stopped working? What happened, and what did you do instead?

Q7 When transferring between iPhone and Android, or phone and a different-OS computer, how do you handle that?

Q8 Describe a time when security/privacy crossed your mind during a file transfer. Did it change your behavior?

Q9 What does it feel like when a transfer fails or arrives corrupted? Tell me about a specific moment.

Theme 3: Motivations, Ideals & Accepted Friction

Goal: Understand retention drivers, unarticulated needs, and normalized workarounds.

Q10 When a file transfer tool works really well, what makes you keep coming back to it?

Q11 Describe your ideal file-sending experience for a scenario that comes up regularly for you.

Q12 What frustrations have you accepted as "the way file transfer works" — things you've stopped trying to fix?

Market Overview

The Fundamental Gap

Existing systems typically optimize for **exactly one transport pathway**. No single consumer app combines all approaches into a unified, adaptive experience.

Cloud Storage
Google Drive
iCloud · Dropbox

Messaging Apps
WeChat
WhatsApp · Telegram

Ecosystem / LAN
AirDrop
Nearby Share

Internet P2P
Send Anywhere
Snapdrop

PAKE Transfer
Magic Wormhole
croc

Competitors analyzed: Google Drive · WeChat · AirDrop · Send Anywhere · Magic Wormhole

Competitor Profiles (1/2)

Google Drive — Cloud Storage

- \+ Persistent storage; cross-device sync; 15 GB free; no recipient install
- − Double bandwidth (sender → server → receiver)
- − Metadata leakage; subscription pressure; unsuited for transient transfer

WeChat — Messaging App

- \+ 1.3B+ users; already in daily use; integrated cloud storage
- − 100 MB limit; aggressive compression; server upload bottleneck
- − Ecosystem lock-in; super-app surveillance risks

AirDrop — Ecosystem LAN Transfer

- \+ Fast (Bluetooth + Wi-Fi Direct); no account; no internet required
- \+ Seamless Apple-to-Apple UX
- − Apple-only; proximity-only; no fallback for remote scenarios
- − Contact-discovery leaks phone numbers and email addresses to passive attackers *(Heinrich et al., 2021)*

Competitor Profiles (2/2)

Send Anywhere — Internet P2P

- \+ Cross-platform (Android, iOS, Web); 6-digit key; no account; 4.3
- − No intelligent routing between LAN/P2P/relay; relay-dependent in most real-world conditions; no resume on failure; no encrypted chat alongside transfer

Magic Wormhole — PAKE Transfer

- \+ Strongest cryptographic model (SPAKE2); relay holds no plaintext; resists offline attack; open-source — *architecturally closest to FluxLink*
- − CLI-only; no mobile app; no LAN fast path (all traffic routed through relay even on same Wi-Fi); no resume; single centralized relay

Comparison Table

Feature	Google Drive	WeChat	AirDrop	Send Anywhere	Magic Wormhole	FluxLink
Cross-platform	✓	Partial	×	✓	✓	✓
LAN / local transfer	×	×	✓	×	×	✓
Internet / relay	✓	✓	×	✓	✓	✓
Auto-routing	×	×	×	×	×	✓
E2EE (all modes)	×	×	Partial	Partial	✓	✓
Encrypted chat	×	×	×	×	×	✓
No account required	✓	×	✓	✓	✓	✓
No ecosystem lock-in	✓	×	×	✓	✓	✓
Mobile app	✓	✓	✓	✓	×	✓
Resume on failure	×	×	×	×	×	✓

Intelligent Transfer Routing Engine

Core Insight

Every competitor optimizes a **single transfer modality** and asks users to manage the rest manually. FluxLink treats the protocol stack as a **scored decision**, not a user configuration.

Priority chain (probed automatically after pairing):

1. **mDNS-based LAN** — lowest latency, highest throughput, no internet dependency
2. **IPv6 direct P2P** — increasingly viable on campus and carrier networks
3. **QUIC/UDP hole-punching** — rendezvous-assisted NAT traversal; no file content on relay
4. **Centralized relay fallback** — guarantees delivery under any condition; relay handles ciphertext only

*The user sees one consistent interaction: **pair, send, done**. Path degradation triggers silent fallback.*

Security Model & Encrypted Chat

Uniform Security Across All Modes

- QR / short-code handshake establishes a session key (SPAKE2)
- All file chunks and chat messages encrypted **before leaving the device**
- Relay server never sees plaintext
- Relay-cached messages cleared weekly

Encrypted Chat Layer

- **No competitor** integrates encrypted chat alongside file transfer
- The conversation about a file is as private as the file itself
- Same E2EE session — no context leakage

Why This is Hard to Copy

The routing engine cannot be added incrementally to a single-modality app. Competitors would need to simultaneously rebuild their network layer, session model, and security architecture — at odds with their current business models and architectural philosophy.

Summary

What We Bring

- Intelligent multi-protocol routing engine
- Uniform E2EE across LAN, P2P, and relay
- Integrated encrypted chat
- No account, no ads, no decision cost

Next Steps

- Conduct user interviews (10–15 participants)
- Synthesize pain points & refine problem framing
- Prototype routing engine core
- Validate pairing UX with paper prototypes

Thank you — Questions?

Thank You